Fluid Flow Through Fluidized Beds

Pre-Lab Plan VI

Unit Operations Lab I

11/14/2019

Group #3, Thursday, PM section

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1. Experimental Objective

The main objective of this experiment is to determine how two different types of particles of relatively small size fluidize. The two particles that we will be working with are sand and resin. Additionally, the pressure drop of the gas (air) will be measured as a function of the gas flow rate. These parameters are measured to obtain the minimum velocity of fluidization and compared it to the results retrieved in lab.

2. Experimental

2.1. Apparatus

Table 1 - Apparatus Equipment Explanations

1	**Batch Fluidized Bed**	Vessel to contain sand or ion exchange resin to illustrate the fluid behavior of air.
2	**DO NOT USE Other Batch Fluidized Bed**	Do not use this fluidized bed and ensure that the valve regulating flow to this fluidized bed is closed during the experiment.
3	**Rotameter 1**	Monitors air flow to the fluidized bed with a max airflow of 36.09 LPM be sure to add to rotameter 2.
4	**Rotameter 2**	Monitors air flow from the main inlet with a max airflow of 426.8 LPM
5	**Pressure Valve and Gauge**	Regulates the main inlet pressure to the system.
6	**Flow Valve and Gauge**	Allows for fine adjustment to the flow of air to the fluidized bed.
7	**Valve**	Allows pressurized air to enter the batch fluidized bed utilized in this laboratory.
8	**Flow Valve**	Adjusts air flow to the fluidized bed for the second rotameter.
9	**Manometer**	Measure pressure differential between inflow air and fluidized bed.
10	**Sand Trap**	Prevents sand and ion exchange resin from entering the pressure differential.

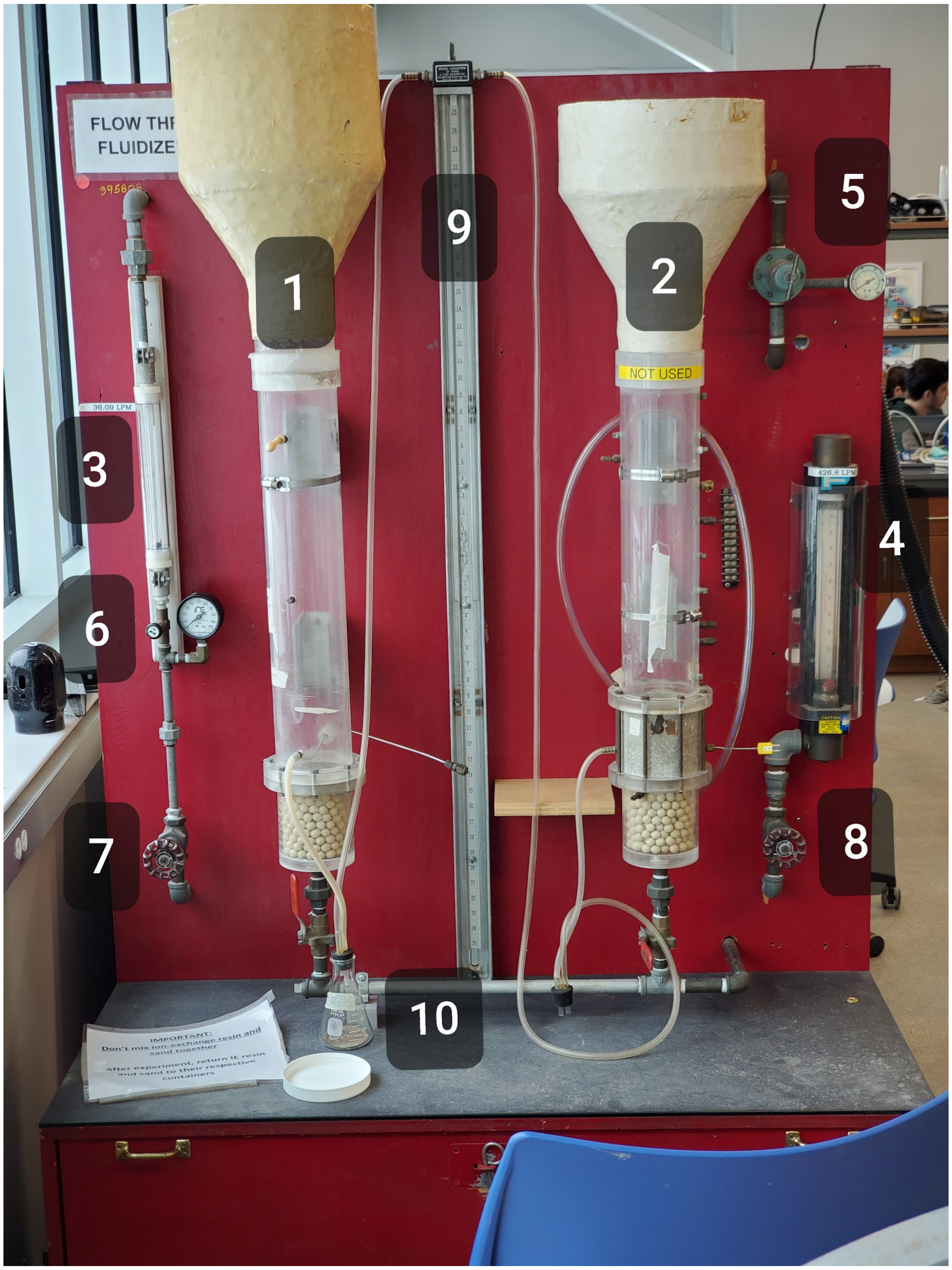

Figure 1 - Process Control System Setup

2.2. Materials and Supplies

Table 2 - Materials and Supplies Descriptions

Material	**Description**
Sand	Fine course material derivative of silica and abundant in amphibious environments.
Ion Exchange Resin	Fine metallic material to highlight changes of the air's fluid behavior through substance of different composition.
Sieve	Utilized to develop a size distribution for the sand based on average particle size.
Funnel	Ensures the safe transition of sand or ion exchange resin particles to the batch fluidized bed
Vacuum	Clean the system of either sand or ion exchange resin
Scale	Mass the expelled sand from the collection of sieves
Graduated Cylinder	Measure the porosity/void fraction

2.3. Experimental Plan

- Fill the vessel to a height of 6" with fine silica sand
- Turn on the air slowly and meanwhile record the pressure drop as a function of gas flow rate
- Increase the flow rate until minimum fluidization conditions are reached

- Decrease the velocity of the gas and while the gas velocity is being decreased to zero, record the pressure drop
- Now fill the bed to 10” with the fine silica sand and repeat
- Clean out the sand and use the vacuum to aid
- Now repeat the experiment with 6” and 10” of ion exchange resin beads

Project Safety Evaluation (10 pts)

Project Title: **Fluid Flow Through Fluidized Beds**

Date: **November 14, 2019**

Hazard	*Yes/No*	Description
Potential electrical hazards?	No	N/A
Potential mechanical hazards?	Yes	Make sure to use valve with the maximum flow rate of 36.09 LPM valve first to ensure that the sand/resin does not receive more flow than needed. When that valve is adjusted first you can then use the 426.8 LPM valve to adjust the flow rate.
Potential pressure hazards?	Yes	The inlet pressure of the air is around 40 psig so make sure you do not adjust the valve next to the pressure gauge. Additionally, keep in mind that pressure gauges may be overpressured and result in injuries.
Potential temperature hazards?	No	N/A
Hazardous/toxic chemicals involved?	Yes	Do not digest the sand or resin that is used. Ion exchange resin can also damage eyesight so use google in laboratory. Also make sure that they do not mix.

Gloves required?	Yes	Regular gloves can be used to perform this experiment.
MSDS reviewed by all team members?	Yes	Sand and Ion exchange Resin MSDS sheet reviewed. For Sand – 14808-60-7 For MBD-15 – 69011-20-7 For Compressed Air – 132259-10-0
Process Parameters	*Max* *Expected*	Possible projectile hazard if pressure vessel is over pressurized and fluid is ejected from the pressure vessel.
Temperature (ºC)	25°C	Air, Sand, and Resin also at room temperature.
Pressure (psig)	40 psig	Inlet air stream supply comes into the system at around 120 psig and gets regulated down to 40 psig.
Safety Controls	Yes	Make sure to be extremely careful with the pressure gauges on the apparatus and to not over pressurize the system since the manometer may break.

I have read relevant background material for the Unit Operations Laboratory entitled: "Process Control: Level Control" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

11/14/2019 _____ Javier Eduardo Mendieta Castillo signature of team member

date

Omar Zahdan ________________

________ 11/14/2019 _______

signature of team member

date

Omar Takrouri __________________

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Joel Finlon ___________________

________ 11/14/2019 _______________

signature of team member

date